



UNIFIED MEMORY (UMA) BUS CONTENTION & QoS ARBITRATION

Preventing High-Bandwidth Video Ingestion DMA from Starving Real-Time Reflex Mailboxes

MASTER 1: CAMERA DMA

MIPI CSI-2 Ingestion Engine

Traffic: Uncompressed Video

Rate: 1.4 GB/s Burst Writes

Payload: 4K @ 60 FPS Pixel Buffer

MASTER 2: NPU ACCEL

Systolic Tensor Array

Traffic: Weight Streaming

Rate: 25 GB/s Read Bursts

Payload: INT8 Filter Tensors

MASTER 3: MPU CPU

ARM Cortex-A Application Cores

Traffic: L3 Cache Refills

Rate: 4.0 GB/s Instruction/Data

Payload: Linux Perception Stack

MASTER 4: MCU MAILBOX

Cortex-M4 Real-Time Safety

Traffic: 1 kHz Safety IPC

Rate: < 1 MB/s Micro-Packets

Requirement: **Jitter < 5 μ s**

DMA Write

Weight Burst

Cache Line

QoS P0

SHARED SYSTEM CROSSBAR & AXI QUALITY-OF-SERVICE (QoS) ARBITER

✖ Unmanaged FIFO Arbitration (Contention Risk):

Unthrottled camera DMA bursts saturate memory channels; NPU weight fetches stall; 1 kHz MCU safety packets queue behind video frames, causing P_{99} latency tail spikes.

✔ Strict AXI QoS Arbitration (Engineered Invariant):

- **Priority 0 (Highest):** Real-time MCU IPC Mailbox (< 2 μ s preemption)
- **Priority 1:** NPU Tensor Weights · **Priority 2 (Paced):** Camera DMA

Unified LPDDR5 DRAM (16 GB · 44 GB/s Peak)

Shared physical pool: OS memory, Model Weights, Camera Frame Buffers

↔ Shared SRAM Mailbox (512 KB · Zero-Wait)

Hardware Mailbox Registers, Lock-Free SPSC Circular Ring Buffers, Atomic Semaphores

🕒 SENSE-TO-ACTUATION LATENCY TAILS UNDER BUS CONTENTION:

⚠ Unmanaged Bus Contention ($P_{99} = 135 \text{ ms} > \tau_{\text{world}}$):



✔ AXI QoS Arbitrated Bus ($P_{99} = 24 \text{ ms} \ll \tau_{\text{world}}$):

